

①

L' Hôpital's Rule for Limits of Form " $\frac{0}{0}$ " or " $\frac{\infty}{\infty}$ "

Assume that f and g are differentiable and that (i) or (ii) holds

(i) $f(x)$ and $g(x)$ both approach 0
as $x \rightarrow a$

(ii) $f(x)$ and $g(x)$ both approach $\pm\infty$
as $x \rightarrow a$

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Assume also that

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = L.$$

Then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L.$

(Here a can be any real number, or ∞ , or $-\infty$. Works for one-sided limits too.)